INSPACE-CAM-2026 Collision Avoidance Filing

Filing Reference ID:	2
Primary Spacecraft:	ISS (ZARYA) (NORAD 25544)
Secondary Spacecraft:	NORAD 900
Operator of Record:	OrbitGuard Swarm Agent Orchestrator
Time of Closest Approach (TCA):	2026-08-25 10:17:19.362061
Filing Status:	Filed
Submission Timestamp:	2026-08-25 15:49:05.198705

Regulatory Briefing & Maneuver Justification

INSPACE-CAM-2026 Collision Avoidance Filing Briefing

REGULATORY STATEMENT FOR SAT-TO-SAT CONJUNCTION MITIGATION

This official filing outlines the emergency collision avoidance maneuver (CAM) planned for ISS (ZARYA) (NORAD 25544) to mitigate a high-risk conjunction with CALSPHERE 1 (NORAD 900).

1. MANEUVER NECESSITY:
Based on high-fidelity orbital propagation and spatial screening, a close approach has been detected with a calculated Time of Closest Approach (TCA) at 2026-08-25T10:17:19.362061. The estimated miss distance is 0.35 km, yielding a Collision Probability (Pc) of 0.000145, which exceeds the standard safety threshold (1.0e-4). This maneuver is critical to safeguard space assets and eliminate collision risk.

2. TARGET SAFETY WINDOW:
The mitigation maneuver will execute approximately 12 hours prior to TCA, establishing a safe physical separation distance exceeding 2.0 km at the moment of closest approach. State vectors will be monitored continuously post-maneuver to verify trajectory stability.

3. IADC SPACE DEBRIS GUIDELINES COMPLIANCE:
All maneuvers are designed to minimize risk to the space environment. Post-maneuver trajectory propagation ensures that the satellite remains within a stable orbit, avoids secondary conjunctions with other cataloged objects, and fully complies with Inter-Agency Space Debris Coordination Committee (IADC) guidelines on space safety and mitigation.

Signed,
OrbitGuard Swarm Agent Orchestrator

Astrometric State Vectors & Covariance Matrices (TCA Epoch)

Below are the exact Earth-Centered Inertial (ECI) position (km) and velocity (km/s) state vectors and uncertainty covariances used to calculate the collision probability, ensuring absolute reproducibility for auditing and regulatory verification.

Parameter	Primary Spacecraft	Secondary Spacecraft
ECI Position Vector (km)	N/A	N/A
ECI Velocity Vector (km/s)	N/A	N/A
ECI Covariance Matrix (km^2)	N/A	N/A

Authorization & Sign-off

Authorized Operator Signature

Date